JIACHENG LIU

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RESEARCH INTERESTS

My research centers on **AI-assisted sensing technologies**, with a particular emphasis on **wearable form factors**. I am passionate about building **activity-detection systems** and **assistive devices** that provide unobtrusive and powerful monitoring and support. I envision a future in which AI genuinely understands human behavior and helps people better understand themselves.

EDUCATION

Cornell University, Ithaca, New York, USA *Aug 2025 – Present*
Ph.D. Student in Information Science
Advisor: Cheng Zhang

Tsinghua University, Beijing, China *Sep 2021 – Jun 2025*
B.Eng. in Creative Design and Intelligent Engineering (CDIE), Xinya College
An interdisciplinary program combining Electrical Engineering and Computer Science (EECS, main part), Mechanical Engineering, and Industrial Design.
GPA: 3.96/4.00, Rank 2 in major.

PUBLICATIONS

** These authors contributed equally to this work.*

Peer-Reviewed and Accepted

[J1] **Jiacheng Liu***, Jiankai Tang*, Guangye Zhao, Ruichen Gui, Songqin Cheng, Taiting Lu, Jian Liu, Weiqiang Wang, Mahanth Gowda, Yuanchun Shi, and Yuntao Wang. 2026. **PPG as a Bridge: Cross-Device Authentication for Smart Wearables with Photoplethysmography**. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT 2026)*. ACM DL

[C2] Ruidong Zhang, **Jiacheng Liu**, François Guimbretière, and Cheng Zhang. 2026. **SoniSpeech: A Large-Scale Open-Vocabulary Tri-Modal Dataset for Wearable Silent Speech Interfaces**. *Conference of the International Speech Communication Association (Interspeech 2026)*. ISCA Archive

[C1] Yu Zhang, Peizhong Gao, Fangzhou Kang, Jiaxiang Li, **Jiacheng Liu**, Qi Lu, and Yingqing Xu. 2024. **OdorAgent: Generate Odor Sequences for Movies Based on Large Language Model**. *2024 IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR 2024)*, pp. 105–114. IEEE Xplore

[W1] Jiankai Tang, Xinyi Li, **Jiacheng Liu**, Xiyuxing Zhang, Zeyu Wang, and Yuntao Wang. 2024. **Camera-Based Remote Physiology Sensing for Hundreds of Subjects Across Skin Tones**. *PhysioCHI: Towards Best Practices for Integrating Physiological Signals in HCI, Workshop at ACM CHI 2024*, Honolulu, HI, USA. arXiv:2404.05003

Preprints and Manuscripts in Preparation

[P1] Ruidong Zhang, **Jiacheng Liu**, Justin Xiang, Ke Li, Babak Sadoughi, Sam Tilsen, François Guimbretière, and Cheng Zhang. **ReVoice: AI-Powered Wearable Acoustic Sensing Restores Natural Voice for Laryngectomees**. *In preparation for the 2027 CHI Conference on Human Factors in Computing Systems (CHI '27).*

[P2] Jiankai Tang*, **Jiacheng Liu***, Renling Tong, Kai Zhu, Zhe Li, Xin Yi, Junliang Xing, Yuanchun Shi, and Yuntao Wang. 2025. **Exploring Reliable PPG Authentication on Smartwatches in Daily Scenarios**. *In preparation for IEEE Transactions on Biometrics, Behavior, and Identity Science (IEEE T-BIOM).* arXiv:2503.23930

[P3] **Jiacheng Liu**, Ruidong Zhang, François Guimbretière, and Cheng Zhang. 2026. **Synthesizing Ultrasonic Echo Profiles from Text for Open-Vocabulary Silent Speech Recognition**. *Submitted to the 2027 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP 2027).* arXiv:2610.15671

RESEARCH EXPERIENCE

Pervasive HCI Group, Tsinghua University *Oct 2023 – Aug 2025*
Undergraduate Research Assistant / Advisor: Prof. Yuntao Wang *Beijing, China*

- Led research on user authentication for wearable devices with photoplethysmography (PPG) and remote PPG (rPPG); delivered two papers in collaboration with Ant Group.
- Contributed to a study on the impact of skin tone diversity on the performance of rPPG models.

SciFi Lab, Cornell University *Jul 2024 – Oct 2024*
Visiting Student / Advisor: Prof. Cheng Zhang *Ithaca, New York, USA*

- Contributed to research on speech synthesis and speech recognition based on active acoustic sensing.
- Studied multimodal alignment and cross-modal learning methods, and their application to speech synthesis and recognition.

The Future Lab, Tsinghua University *Mar 2023 – Sep 2023*
Undergraduate Research Assistant / Advisor: Prof. Yingqing Xu *Beijing, China*

- Contributed to research on novel interactions for olfactory experiences in virtual reality.
- Built a zero-shot olfactory feedback system for VR films based on the CLIP vision-language model.

INDUSTRY EXPERIENCE

BehaviorAI *Jun 2026 – Aug 2026*
AI Engineer Intern, founding team *New York City, New York, USA*

- Built machine learning models for human behavior tracking from wearable sensor data.
- Ran signal analysis and quality control over the collected data, gating what entered model training.
- Provided technical support for data collection, including the design and implementation of the in-house collection software.

TEACHING EXPERIENCE

INFO 4340/5440 – App Design and Prototyping *Spring 2026*
Teaching Assistant / Instructor: Prof. Kyle Harms *Ithaca, New York, USA*

INFO 1300 – Introductory Design and Programming for the Web *Fall 2025*
Teaching Assistant / Instructor: Prof. Kyle Harms *Ithaca, New York, USA*

PROFESSIONAL SERVICE

Reviewer
- ACM Conference on Human Factors in Computing Systems (CHI) *2026*
- Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT) *2026*

HONORS AND FUNDING

Awards and Scholarships
- Outstanding Undergraduate Thesis, Tsinghua University (sole recipient in cohort) *Jul. 2025*
- Xinya College Academic Excellence Award *Jul. 2025*
- Tsinghua Friendship – Luo Yuehua Scholarship (3 recipients college-wide) *Nov. 2024*
- Tsinghua Friendship – Toyota Scholarship (sole recipient college-wide) *Oct. 2023*
- Passion for Reading Scholarship (2 recipients college-wide) *Oct. 2022*

Research Funding
- Tsinghua University Academic Advancement Plan *Oct. 2024*
- Beijing Natural Science Foundation *Oct. 2023*

SKILLS

Programming:	Python, C/C++, Swift, MATLAB, JavaScript, Shell
Modeling Experience:	Hand pose tracking, body pose tracking, speech synthesis and recognition, identity authentication
Hardware & Fabrication:	Arduino, MicroPython, Verilog, SolidWorks, Fusion 360, digital fabrication
Design & Front-End:	Figma, HTML/CSS, Xcode